# Rahil Ehtesham Mehr

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## Education

**Sharif University of Technology** Sep. 2022 – Present *Expected Jun. 2027*
Bachelor of Science in Physics *Tehran, Iran*
***GPA (Last 2 Years): 17.44/20***
***Overall GPA: 15.98/20*** *(Top third of class)*
*Top-ranked university in Iran, according to QS ranking 2025*

## Research Interests

- Experimental Condensed Matter Physics
- Perovskite Solar Cells
- Nanoscale Materials and Devices
- Two-Dimensional Materials

## Research Experience

**Nanoparticles and Coatings Lab (NCL)** Jun. 2026 – Present
Supervisor: Dr.Nima Taghavinia | Perovskite Solar Cells • Thin Film Deposition *Sharif University of Technology*
- Working on perovskite-based solar cells, with hands-on experience in various thin-film deposition techniques for photovoltaic layer fabrication

**Gold-Assisted Exfoliation of Large-Area Monolayer TMDs** Dec. 2024 – Present
Supervisor: Dr. Ali Khademi | 2D Materials • Nanofabrication *Sharif University of Technology*
- Established a gold-assisted exfoliation technique for producing large-area, single-crystal TMD monolayers by leveraging the strong Au–chalcogen adhesion to overcome interlayer van der Waals bonding. This approach enables deterministic, high-yield exfoliation of macroscopic monolayers with quality comparable to Scotch-tape exfoliation, suitable for quantum materials and device applications.

**URA – Non-Invasive Laser–Tissue Interaction Simulation** Jun. 2024 – Jun. 2025
Supervisor: Dr. Rasoul Sedighi | Biomedical Photonics • Optics *Sharif University of Technology*
- Developed computational models of optical and thermal interactions between coherent light and biological tissue for non-invasive diagnostic applications
- Performed photon transport simulations in multi-layered skin models to optimize clinical parameters and enhance understanding of laser-based treatments

## Honors & Awards

**Professional Chess Player** 2010 - 2020
♟ FIDE Rating: 1850
🏅 Over 60 regional and national titles

**Member of the University Chess Team** Summer 2024 - Present
2024 National University Sports Olympiad: 2nd place Sharif University of Technology
2024 Regional Tournament: 1st place (Individual) - 2nd place (Team)

**National University Entrance Exam (Concours)** Fall 2022
Ranked in the top 1% in the field of Math and Physics

## Teaching Experience

**Vacuum Technology** (TA) Fall 2026
*Instructor: Dr. Omid Akhavan* *Sharif University of Technology*
Designing & Grading Homework Assignments | Holding Weekly Problem-Solving Sessions

**Chess Instructor**
*Private Group Tutoring*
Taught chess fundamentals to elementary students, developing age-appropriate curriculum covering basic tactics, strategic concepts, and competitive play while fostering critical thinking and problem-solving skills.

## Remarkable Projects

**High-Vacuum System Design** Fall 2024

*Supervisor: Prof. Ali Esfandiar*

This project focuses on the design, optimization, and testing of a high-vacuum system used in SEM and XPS instruments. The system aims to achieve ultra-high vacuum conditions to enhance the accuracy and resolution of nanoscale surface analysis.

**Investigating Methods to Increase the Critical Temperature of Superconductors** Spring 2024

*Supervisor: Prof. Shant Baghram*

Explored and analyzed methods for raising the critical temperature in high-temperature superconductors to advance their practical applications

**Mechanical Constraint Effect on DNA Persistence Length** Spring 2025

*Supervisor: Prof. Seyed-Nader Seyed Reihani*

Quantified how mechanical confinement and tensile forces alter the apparent persistence length of DNA. The project involved analyzing the deviation from Worm-Like Chain (WLC) model predictions under constrained conditions, relevant to genomic packaging and single-molecule biophysics

**Mapping the Universe through the Cosmic Microwave Background** Spring 2023

*Supervisors: Prof. Baharm Mashhoon, Farhad Ardalan*

Examines how CMB data provide a snapshot of primordial density fluctuations shaping today's cosmos.

## Technical Skills

**Data Analysis and Programming Languages**: Python, Excel, LaTex

**Thin-Film Deposition Techniques**: Spin Coating, Blade Coating

**Characterization Methods**: Familiar with SEM, XPS System operation

**Other Skills**: Team Collaboration, Cleanroom Experience, Sample Preparation, Vacuum System Operation

## Language Proficiency

**English**: TOEFL iBT 110/120 (5.5/6), CEFR C1

**Persian**: Native

## Notable Courses

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|---|---|---|---|
| **Thin Films** | 20/20 | **Physics 4** | 18/20 |
| **Vacuum Technology** | 17.7/20 | **Physics 4 Lab** | 19.2/20 |
| **Electromagnetics** | 18/20 | **Electronic** | 19/20 |
| **Biophysics** | 17/20 | **Statistical Mechanics** | 20/20 |

## Volunteering & Service

**Interdepartmental Chess Tournament Director & Organizer** May 2024 & 2025

*Sharif University of Technology Physics Department*

• Managed all event logistics and administrative tasks, successfully engaging faculty members and students.

**Chess Puzzle Columnist** Sept. 2025 – Present

*Sharif University Newspaper*

## Extra Curricular Activities

**Competitive chess player** on Lichess whenever I'm not working; Staying active through **Running, Skating, and Karate**; **Cinema enthusiast**; **Tea connoisseur**

## References

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